

Identification of Undocumented Meat Samples Seized at Heathrow Airport

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Scenario

A suitcase containing several packages of unidentified frozen meat has been intercepted at Heathrow Airport by UK Border Force officers during routine inspections. The shipment had no import documentation, raising concerns that the meat may originate from protected or endangered species.

Given the serious implications for public health, food safety, and wildlife conservation, the case has been referred to the Animal and Plant Health Agency (APHA). As a bioinformatician assisting with the inquiry, you have been tasked with determining the species of origin for each meat sample. DNA barcoding has already been performed in the laboratory, and you have now received the resulting FASTAQ sequence files for analysis.

Abstract

A modular pipeline was constructed to take the provided sequence data (`samples.zip`) through to the visualisation of a phylogenetic tree for sample identification. The minimal input design of the pipeline and provided `environment.yml` file maximises reproducibility and minimises incorporating bias or human error into the analysis

Introduction

Rarity status attributed to a species often increases its price and demand, with critically endangered species fetching the highest prices [1, 2]. The fallout of this is a positive feedback loop between rarity and demand which may drive an endangered species towards extinction and threaten whole ecosystems [3]. Species-level identification of undocumented meat samples is key in public health for validating the meat being delivered to consumers, but may also uncover networks carrying out illegal imports of endangered species, thus aiding conservation efforts by providing routes to intervene in the trade of these species.

If the morphology of a sample is compromised, DNA barcoding - comparing genetic sequences in a sample against known references - can aid in resolving the species of the sample [4]. Mitochondrial DNA (mtDNA) genes are commonly used for DNA barcoding because they occur in high copy numbers relative to nuclear DNA and typically lack recombination, keeping

within-species diversity low which enables species-level identification [5]. Cytochrome c oxidase I (COI), found within mtDNA, is the primary barcode marker in the animal kingdom, and reference sequences for COI are abundant in databases such as BOLD and NCBI, permitting barcoding against a diverse set of organisms [6, 7].

This analysis aims to resolve the species of four meat samples intercepted by UK Border Force officers at Heathrow Airport. The sequenced samples provide the starting point of this analysis pipeline.

Methods

Overview

A modular pipeline was constructed to take the provided sequence data (`samples.zip`) through to the visualisation of a phylogenetic tree for sample identification. The four key pipeline stages are described here.

QC and Pre-Processing

FASTQ sequence files were first extracted from `samples.zip` before undergoing quality control to enforce a consistent FASTQ structure - including headers and line counts. Every part file (1-3) belonging to the four samples (A-D) were then concatenated into multi-read FASTQ files by sample. `seqtk` [8] was used to convert each to FASTA while masking any bases with quality scores < 20 to 'N', ensuring bases were called at a confidence $\geq 99\%$ during sequencing to reduce the chances of false positives and increase confidence in downstream BLAST results.

BLAST and FASTA Concatenation

BLAST [9] searches were carried out using the FASTA files as query sequences against the NCBI Nucleotide database using `blastn` [10]. As the samples were labeled parts 1-3 without information on genomic region, each part was treated as a separate fragment and BLAST results for each sample were split into part-specific files, enabling downstream alignment per-part across all samples. These results were then sorted by ascending e-value, descending bitscore, and descending alignment length before collapsing by species with `taxonkit` [11]. The top four for each part and sample were selected.

`efetch` [12] was used to download the FASTA sequences for each BLAST hit accession, and the `sstart` and `ssend` values were used to trim these sequences to match the query region using `seqtk`, reverse complementing if an opposite strand is matched in BLAST. COI sequences of three *Selachii* species were retrieved using `efetch` as an outgroup before all FASTA sequences were concatenated by part into three complete FASTA files (parts 1-3) using `seqkit` [13].

Translation and Alignment

Translation of the FASTA sequences using the vertebrate mitochondrial code was carried out using `Biopython` [14]. To counteract potential frame shift errors during translation, ORFs were

identified and selected as the protein sequence for each read.

MUSCLE5 [15] was used to carry out multiple sequence alignments on the three part-protein files, and trimAl [16] removed poorly aligned regions while retaining internal sites for a more stable tree build. Inspection of BLAST results indicated that each part derived from COI, so the trimmed alignments were concatenated into a supermatrix alignment with catfasta2phym1 [17], retaining partitions for each part.

Tree Building and Viewing

Phylogenetic inference was performed using maximum likelihood in IQ-TREE3 [18] with ModelFinder Plus selecting the best model to improve accuracy, and included 1000 ultrafast bootstrap replicates to evaluate branch support. The Python library ete3 [19] was used to visually display the phylogenetic tree as well as rooting to the outgroup *Selachii* clade.

Results

BLAST results provided preliminary identifications for each sample. Sample A consistently matched Right and Bowhead Whales (*Balaenidae*), with sample B returning Beluga whale and Narwhal (*Monodontidae*). Sample C matched true tunas (*Thunnus* spp.), and sample D matched sea turtle species (*Cryptodira*) as well as *Homo sapiens*. Top species per part are shown in Table 1.

Table 1: Top BLAST hit per part across all samples

Sample	Part	Species	% Identity	e-value
A	1	<i>Eubalaena japonica</i>	97.5	0.0
A	2	<i>Eubalaena japonica</i>	99.6	0.0
A	3	<i>Eubalaena japonica</i>	93.4	7.0e-160
B	1	<i>Delphinapterus leucas</i>	92.3	0.0
B	2	<i>Delphinapterus leucas</i>	92.8	0.0
B	3	<i>Monodon monoceros</i>	89.2	0.0
C	1	<i>Thunnus thynnus</i>	81.7	3.7e-103
C	2	—	—	—
C	3	<i>Thunnus thynnus</i>	89.2	1.2e-157
D	1	<i>Dermochelys coriacea</i>	98.3	0.0
D	2	<i>Dermochelys coriacea</i>	96.2	0.0
D	3	<i>Homo Sapiens</i>	100	6.3e-46

The supermatrix alignment of the translated protein sequences shows high overall homology across all sequences, with good sequence coverage in all except sample C and related reference sequences (Fig. 1). Sample D also displays a potential frameshift within a highly conserved region at the tail-end of part 1.

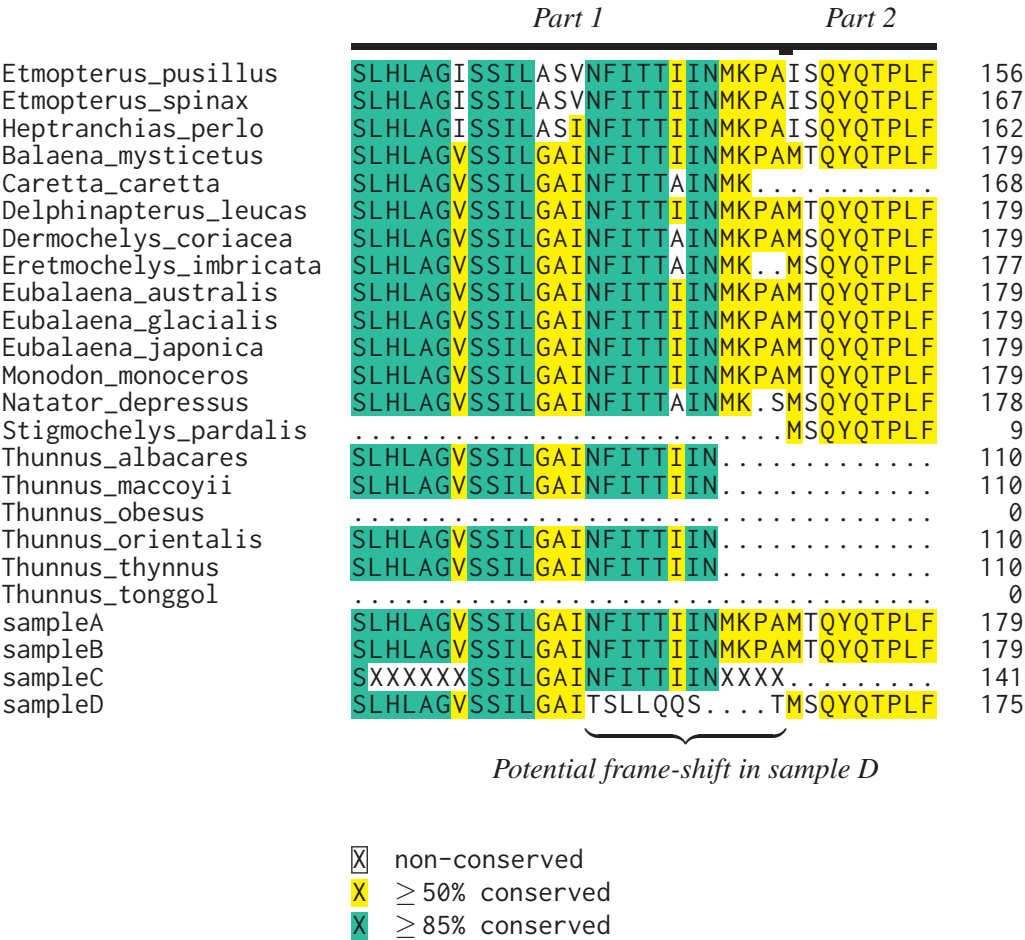


Figure 1: Window of supermatrix alignment with potential frame-shift error in sampleD near the highly-conserved NFITTIIN region.

The resulting phylogenetic tree (Fig. 2) recovered four well resolved vertebrate clades: a Cetacea (whales) clade, containing samples A and B; a Testudine (turtle and tortoises) clade, containing sample D; a Scombridae (tuna) clade, containing sample C; and the outgroup Selachii (sharks) clade. Bootstrap support at nodes placing each sample within each major clade was $\geq 90\%$, indicating consistently supported sample placement within Cetacea, Testudines, and Scombridae.

Within Cetacea, sample A fell into a strongly supported Balaenid clade (96% bootstrap support), and within the Balaenid clade there was weak support for placement of sample A with the

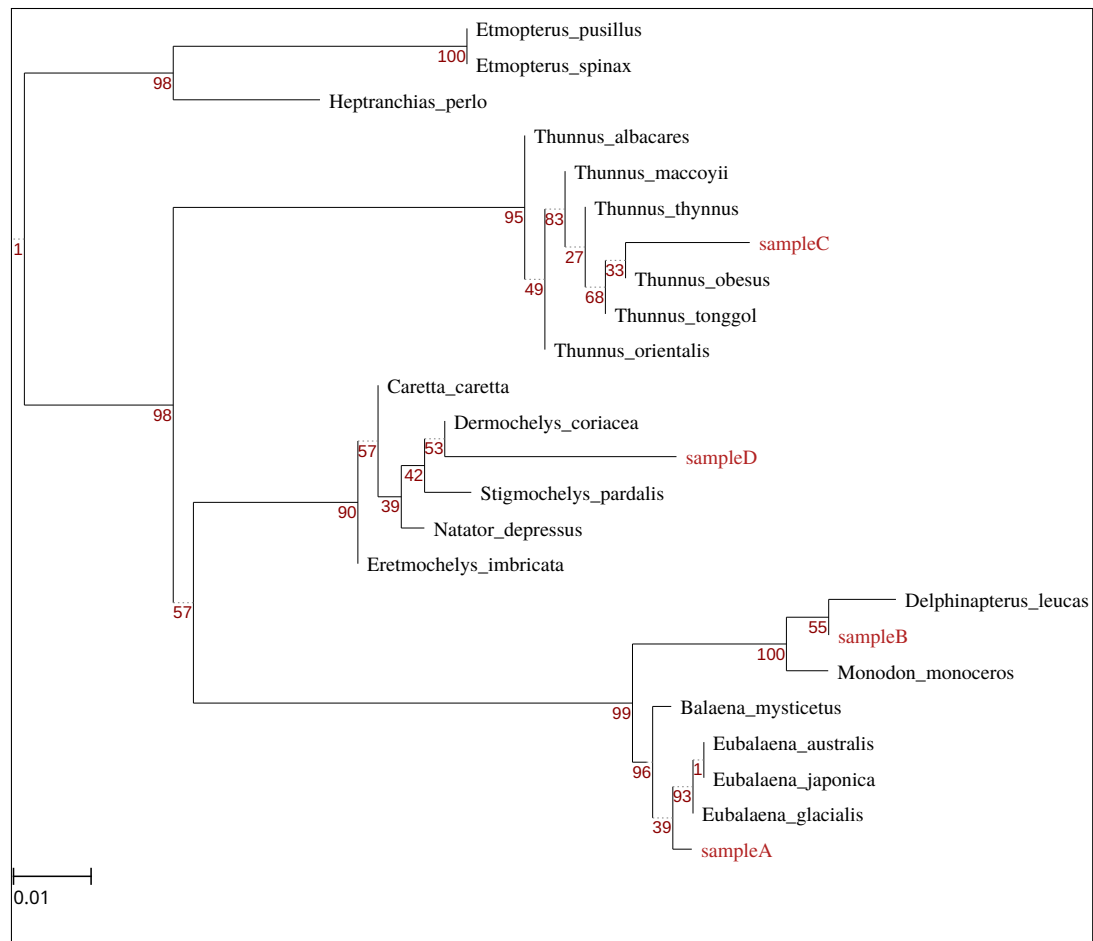


Figure 2: Maximum likelihood phylogenetic tree with placement of unknown meat samples and top BLAST hits.

Discussion

Where cyt b and D-Loop sequences do exist, very few of these have been generated as part of forensic studies [23], with the vast majority submitted to databases as part of phylogenetic, molecular ecological and conservation studies [24], [25], [26]. Such sequence data can be generated and accepted into databases without any need for standard protocols or quality control, raising doubts over their suitability for forensic application [5]

However, unlike BOLD, GenBank does not store sequence chromatograms, collection metadata or photographs [6]. BOLD is a curation tool that also stores sequences, while GenBank is just a sequence repository. [20]

As pointed out by Efron, Halloran and Holmes (1996), the bootstrap compares the bootstrapped values to the original estimate, not to the truth. The essential bootstrap identity is that the bootstrapped values are to the original estimate what the original estimate is to the true parameter [21]

References

- [1] Yik-Hei Sung and Jonathan J Fong. Assessing consumer trends and illegal activity by monitoring the online wildlife trade. *Biological conservation*, 227:219–225, 2018.
- [2] Franck Courchamp, Elena Angulo, Philippe Rivalan, Richard J Hall, Laetitia Signoret, Leigh Bull, and Yves Meinard. Rarity value and species extinction: the anthropogenic allee effect. *PLoS biology*, 4(12):e415, 2006.
- [3] Liam J Hughes, Oscar Morton, Brett R Scheffers, and David P Edwards. The ecological drivers and consequences of wildlife trade. *Biological Reviews*, 98(3):775–791, 2023.
- [4] Pilar A Haye, Nicolás I Segovia, Raúl Vera, María de los Ángeles Gallardo, and Cristian Gallardo-Escárate. Authentication of commercialized crab-meat in chile using dna barcoding. *Food Control*, 25(1):239–244, 2012.
- [5] Nick Dawnay, Rob Ogden, Ross McEwing, Gary R Carvalho, and Roger S Thorpe. Validation of the barcoding gene coi for use in forensic genetic species identification. *Forensic science international*, 173(1):1–6, 2007.
- [6] Teresita M Porter and Mehrdad Hajibabaei. Over 2.5 million coi sequences in genbank and growing. *PloS one*, 13(9):e0200177, 2018.
- [7] Sujeevan Ratnasingham, Catherine Wei, Dean Chan, Jireh Agda, Josh Agda, Liliana Ballesteros-Mejia, Hamza Ait Boutou, Zak Mohammad El Bastami, Eddie Ma, Ramya Manjunath, et al. Bold v4: A centralized bioinformatics platform for dna-based biodiversity data. In *DNA barcoding: Methods and protocols*, pages 403–441. Springer, 2024.
- [8] Heng Li. seqtk: Toolkit for processing sequences in FASTA/Q formats . <https://github.com/lh3/seqtk>, 2012.
- [9] Stephen F Altschul, Warren Gish, Webb Miller, Eugene W Myers, and David J Lipman. Basic local alignment search tool. *Journal of molecular biology*, 215(3):403–410, 1990.
- [10] Christiam Camacho, George Coulouris, Vahram Avagyan, Ning Ma, Jason Papadopoulos, Kevin Bealer, and Thomas L Madden. Blast+: architecture and applications. *BMC bioinformatics*, 10(1):421, 2009.
- [11] Wei Shen and Hong Ren. Taxonkit: A practical and efficient ncbi taxonomy toolkit. *Journal of genetics and genomics*, 48(9):844–850, 2021.
- [12] Jonathan Kans. Entrez direct: E-utilities on the unix command line. In *Entrez programming utilities help [Internet]*. National Center for Biotechnology Information (US), 2024.
- [13] Wei Shen, Shuai Le, Yan Li, and Fuquan Hu. Seqkit: a cross-platform and ultrafast toolkit for fasta/q file manipulation. *PloS one*, 11(10):e0163962, 2016.
- [14] Peter JA Cock, Tiago Antao, Jeffrey T Chang, Brad A Chapman, Cymon J Cox, Andrew Dalke, Iddo Friedberg, Thomas Hamelryck, Frank Kauff, Bartek Wilczynski, et al. Biopython: freely available python tools for computational molecular biology and bioinformatics. *Bioinformatics*, 25(11):1422, 2009.
- [15] Robert C Edgar. Muscle5: High-accuracy alignment ensembles enable unbiased assessments of sequence homology and phylogeny. *Nature communications*, 13(1):6968, 2022.

- [16] Salvador Capella-Gutiérrez, José M Silla-Martínez, and Toni Gabaldón. trimal: a tool for automated alignment trimming in large-scale phylogenetic analyses. *Bioinformatics*, 25(15):1972–1973, 2009.
- [17] Johan Nylander. catfasta2phym1 – Concatenate FASTA alignments to PHYL, PHYLIP, or FASTA format. <https://github.com/nylander/catfasta2phym1>, 2010.
- [18] Thomas KF Wong, Nhan Ly-Trong, Huaiyan Ren, Hector Baños, Andrew J Roger, Edward Susko, Chris Bielow, Nicola De Maio, Nick Goldman, Matthew W Hahn, et al. Iq-tree 3: Phylogenomic inference software using complex evolutionary models. 2025.
- [19] Jaime Huerta-Cepas, François Serra, and Peer Bork. Ete 3: reconstruction, analysis, and visualization of phylogenomic data. *Molecular biology and evolution*, 33(6):1635–1638, 2016.
- [20] Kelly A Meiklejohn, Natalie Damaso, and James M Robertson. Assessment of bold and genbank—their accuracy and reliability for the identification of biological materials. *PloS one*, 14(6):e0217084, 2019.
- [21] Susan Holmes. Bootstrapping phylogenetic trees: theory and methods. *Statistical Science*, 18(2): 241–255, 2003.

Supplementary Data

Supplementary data and all scripts can be found within the GitHub repository for this project:

https://github.com/archiebenn/BIOLM0051_analysis.git

Appendix: Exploratory Figures in Suspected Sequencing Error



Figure 3: Needleman-Wunsch global pairwise alignment at the tail end of sampleD_part1 against closest reference sequence. Alignment gap indicates a possible base miss in sampleD_part1 during sequencing.

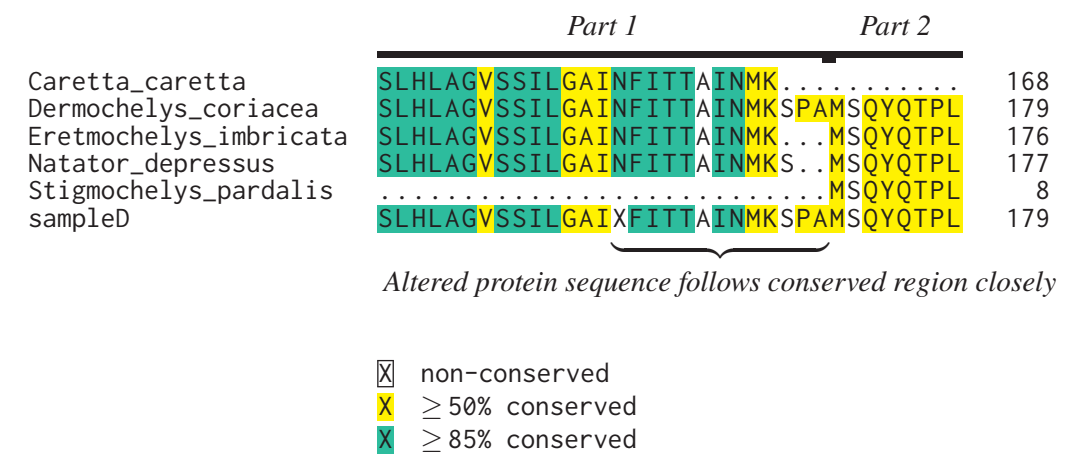


Figure 4: Subset of supermatrix alignment with sampleD and inner clade after inserting one ambiguous base (N) to the nucleotide sequence of sampleD_part1. This base resolves the apparent frame-shift, resulting in a protein sequence which aligns much more closely to highly-conserved NFITTIIN region compared to in Fig. 1.

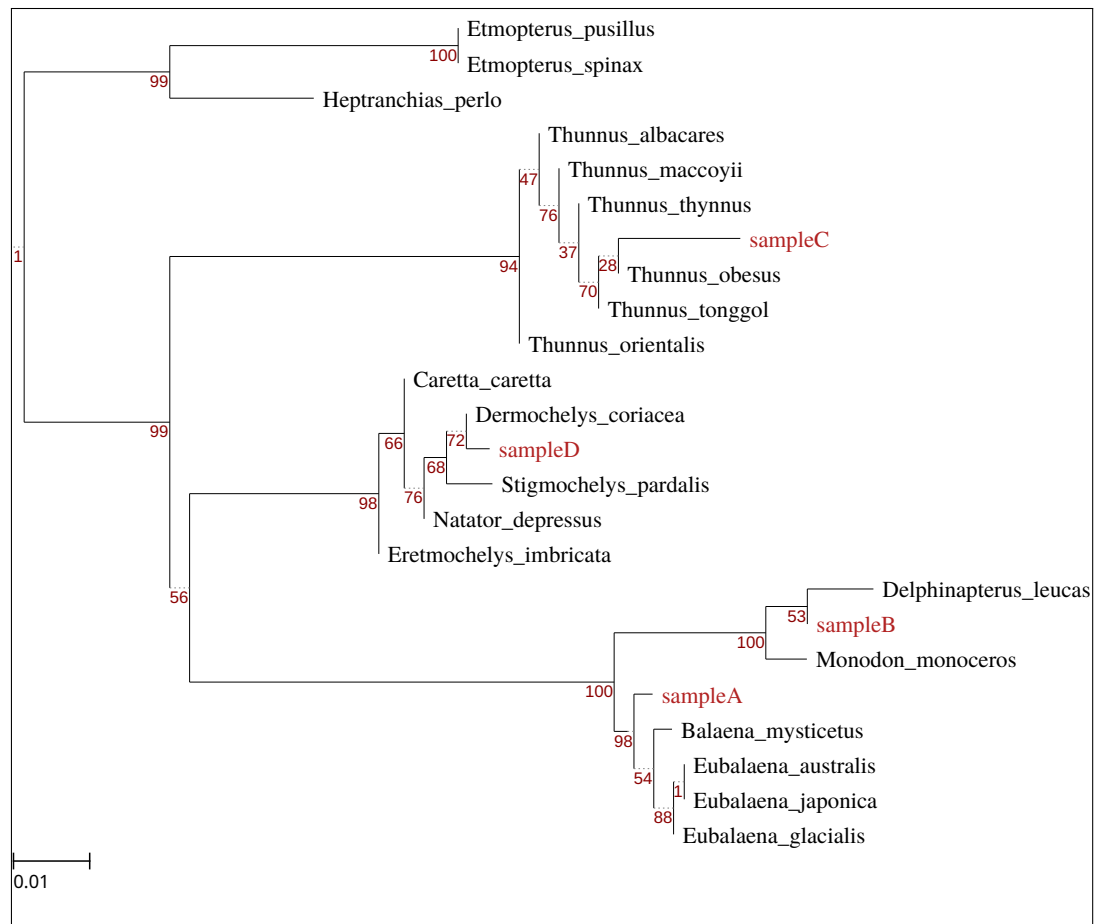


Figure 5: Maximum-likelihood phylogenetic tree after manually corrected alignment in which one ambiguous base (N) was inserted into sampleD.part1. This tree is shown to demonstrate the impact of sequencing error and downstream frame-shift on branch lengths compared to Fig. 2.